

Unit VI: Optics

Chapter–9: Ray Optics and Optical Instruments

GIST OF THE CHAPTER:

Ray Optics: Reflection of light, spherical mirrors, mirror formula, refraction of light, total internal reflection and optical fibers, refraction at spherical surfaces, lenses, thin lens formula, lens maker's formula, magnification, power of a lens, combination of thin lenses in contact, refraction of light through a prism.

Optical instruments: Microscopes and astronomical telescopes (reflecting and refracting) and their magnifying powers.

CONTENT/ CONCEPTS-

Reflection

Reflection is the phenomenon of changing the path of light without any change in the medium.

Reflection of Light

The returning back of light in the same medium from which it has come after striking a surface is called reflection of light.

Laws of Reflection

Two laws of reflection are given as below:

- (i) The angle of incidence i is equal to the angle of reflection r . i.e. $\angle i = \angle r$.
- (ii) The incident ray, reflected ray and normal to the reflecting surface at the point of incidence all lie in the same plane.

A SPHERICAL MIRROR:

A spherical mirror whose reflecting surface is curved inwards, i.e. faces towards the centre of the mirror, is called a concave mirror

A spherical mirror whose reflecting surface is curved outwards, i.e. faces away from the centre of the mirror, and is called a convex mirror.

Pole (P) is the centre of reflecting surface lying on the surface.

Centre of curvature (C) is the centre of the imaginary sphere from which spherical mirror is cut out.

Radius of curvature (R) is the distance between the pole and the centre of curvature.

Principal axis (PCX or CPX) is the line passing through the pole and the centre of curvature and extending to ∞ . It is the normal to the mirror at the pole.

Principal Focus (F) is the point on the principal axis at which the incident rays of light parallel to principal axis either really pass through or appear to pass through after getting reflected from the mirror.

Focal length (f) is the distance between the pole and the principal focus.

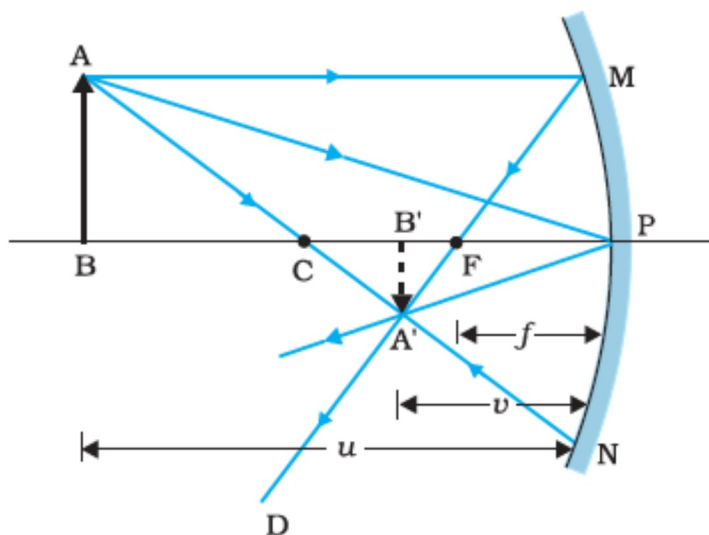
MIRROR FORMULA

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

u – Object distance

v – Image distance

f – Focal length of the mirror



MAGNIFICATION PRODUCED BY A MIRROR

Magnification produced by a mirror is defined as the ratio of the size of the image to the size of the object.

Magnification produced by a mirror is also defined as the ratio of the image distance to object distance.

$$m = \frac{h'}{h} = -\frac{v}{u}$$

SIGN CONVENTIONS FOR REFLECTION BY SPHERICAL MIRRORS

1. The object is always placed to the left of the mirror. i.e. the incident rays from the object always move from left to right.
2. All distances parallel to the principal axis are measured from the pole (P) of the mirror.
3. All the distances measured to the right of the Pole (along +ve x-axis) are taken +ve while those measured to the left of the Pole (along -ve x-axis) are taken -ve.
4. Distances measured perpendicular to and above the principal axis (along +ve y-axis) are taken +ve while those measured below the principal axis (along -ve y-axis) are taken -ve.

TOTAL INTERNAL REFLECTION (TIR)

Total Internal Reflection is the phenomenon of complete reflection of light back into the same medium for angles of incidence greater than the critical angle of that medium.

Conditions for TIR:

The incident ray must be in optically denser medium.

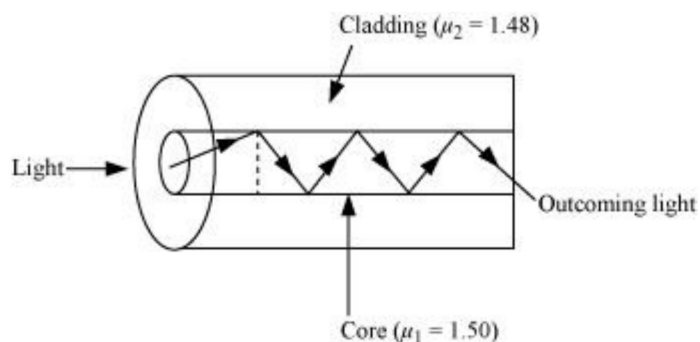
The angle of incidence in the denser medium must be greater than the critical angle for the pair of media in contact.

$$i > i_c \quad \text{and} \quad \text{refractive index } n = \frac{1}{\sin i_c}$$

OPTICAL FIBRE

Optical fibres consist of several thousands of very long fine quality fibres of glass or quartz.

The diameter of each fibre is of the order of 10^{-4} cm with refractive index of material being of the order of 1.5.



The fibres are coated with a thin layer of material of lower refractive index of the order of 1.48.

Propagation of light through an optical fibre:

Light incident on one end of the fibre at a small angle passes inside and undergoes repeated total internal reflections inside the fibre. It finally comes out of the other end, even if the fibre is bent or twisted in any form.

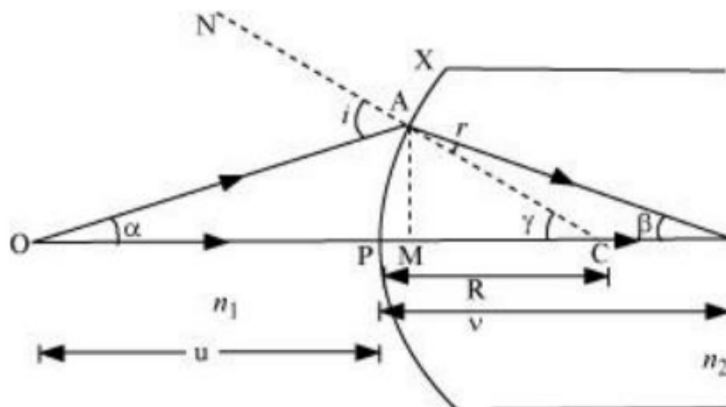
REFRACTION THROUGH A SPHERICAL SURFACE

Consider a point object O lying on the principal axis of the surface.

From A , draw $AM \perp OI$

Let $\angle AOM = \alpha$, $\angle AIM = \beta$, $\angle ACM = \gamma$

As external angle of a triangle is equal to



sum of internal opposite angles,

Therefore, in ΔIAC , $r + \beta = \gamma \Rightarrow r = \gamma - \beta$ — — — — (i)

Similarly, in ΔOBC , $i = \alpha + \gamma$ — — — — (ii)

According to Snell's law,

$$\frac{n_2}{n_1} = \frac{\sin i}{\sin r} = \frac{i}{r} \quad (2)$$

($\because$ Angles are small) $\Rightarrow n_2 r = n_1 i$

Using (i) & (ii), we obtain $\Rightarrow n_2(\gamma - \beta) = n_1(\alpha + \gamma)$

As angle α , β , and γ are small, using $\tan \theta \approx \theta$, we obtain

$$\therefore n_1 \left(\frac{AM}{MO} + \frac{AM}{MC} \right) = n_2 \left(\frac{AM}{MC} - \frac{AM}{MI} \right) \quad (3)$$

As aperture of the spherical surface is small, M is close to P.

Therefore using new Cartesian sign conventions $MO \approx PO = -u$, $MI \approx PI = +v$, $MC \approx PC = R$

From (3),

$$\Rightarrow n_2 \left(\frac{1}{PC} - \frac{1}{PI} \right) = n_1 \left(\frac{1}{PO} - \frac{1}{PC} \right) \Rightarrow \frac{n_2}{PI} + \frac{n_1}{PO} = \frac{n_2 - n_1}{PC}$$

Using new Cartesian sign conventions, we put

$$\Rightarrow \frac{n_2}{v} + \frac{n_1}{-u} = \frac{n_2 - n_1}{R}$$

(i) Refraction from rarer to denser medium

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R} \quad (n_2 > n_1)$$

(ii) Refraction from denser to rarer medium

$$\frac{n_1}{v} - \frac{n_2}{u} = \frac{n_1 - n_2}{R} \quad (n_1 > n_2)$$

Refraction by Spherical Lenses

Lenses whose refracting surfaces are spherical are called ‘spherical lenses’.

A spherical lens whose refracting surfaces are bulging outwards at the centre is called a ‘double convex lens’. It is thicker in the middle compared to the edges.

A spherical lens whose refracting surfaces are curved inwards at the centre is called a ‘double concave lens’. It is thinner in the middle compared to the edges.

LENS MAKER’S FORMULA

- (i) Refraction from rarer to denser medium

$$\frac{n_2}{v_1} - \frac{n_1}{u} = \frac{n_2 - n_1}{R_1}$$

$$(n_2 > n_1)$$

- (ii) Refraction from denser to rarer medium

$$\frac{n_1}{v} - \frac{n_2}{v_1} = \frac{n_1 - n_2}{R_2}$$

$$(n_1 > n_2)$$

Adding two equations

$$\frac{n_1}{v} - \frac{n_1}{u} = (n_2 - n_1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

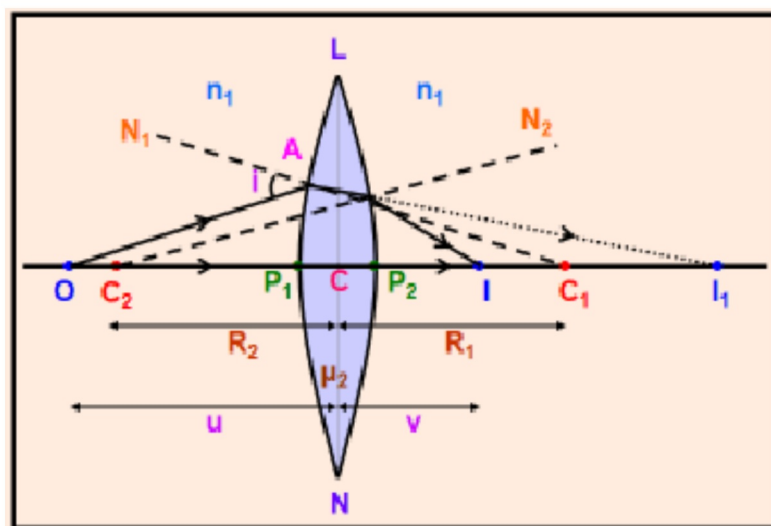
Taking $v = f$ when $u = \infty$

$$\frac{1}{f} = \left(\frac{n_2}{n_1} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

LENS FORMULA

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

LINEAR MAGNIFICATION



Linear magnification produced by a lens is defined as the ratio of the size of the image to the size of the object.

$$m = \frac{h'}{h} = \frac{v}{u}$$

POWER OF A LENS

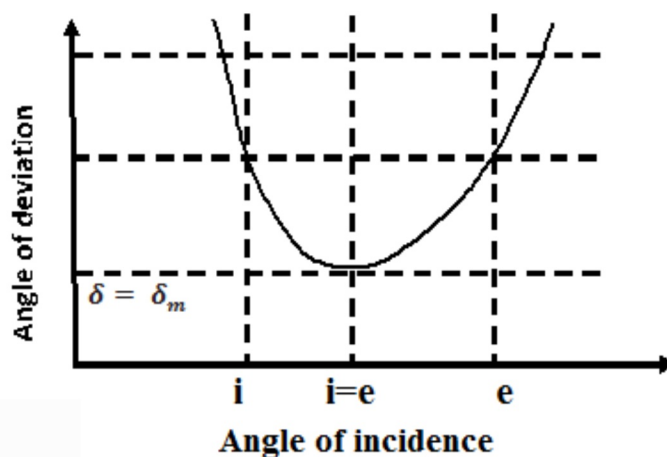
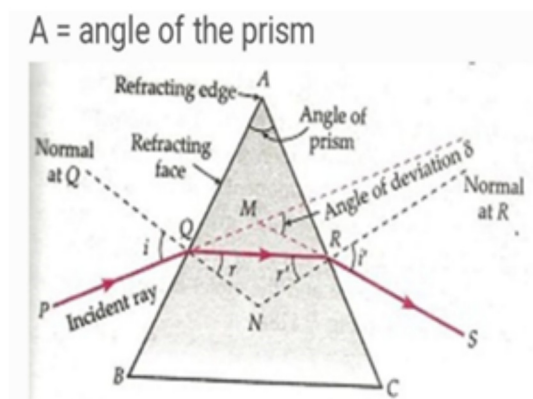
Power of a lens is its ability to bend a ray of light falling on it and is reciprocal of its focal length.

When f is in metre, power is measured in Dioptre (D).

$$P = \frac{1}{f}$$

REFRACTIVE INDEX OF THE MATERIAL OF A PRISM

Refraction by a Small-angled Prism for Small angle of Incidence:



In prism $\Rightarrow r_1 + r_2 = A$ And The total deviation $= \delta = i + e - A$

In general, any given value of δ , corresponds to two values i and hence of e . This is related to the fact that the path of ray can be traced back, resulting in the same angle of deviation.

When $\delta = \delta_m$, $\Rightarrow i = e$ and $\Rightarrow r_1 = r_2 = r$ (say) $\Rightarrow r = \frac{A}{2}$

and $\Rightarrow \delta_m = 2i - A \Rightarrow i = \frac{\delta_m + A}{2}$

$$\text{PRISM FORMULA} \quad n = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)} \quad \text{for thin prism } \delta_m = A(n - 1)$$

SIMPLE MICROSCOPE

It is a convex lens of short focal length

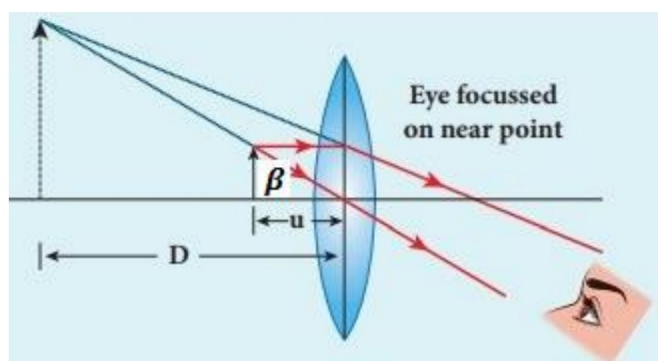
Magnifying power (m): The magnifying power of a microscope is defined as the ratio of the angle β subtended by the image at the eye to the angle α subtended by the object seen directly, when both lie at the least distance of distinct vision.

$$m = \frac{\beta}{\alpha} = \frac{h/u}{h/D} = \frac{D}{u} = \frac{D}{v} + \frac{D}{f}$$

(using lense formula with sign)

For image at near point ($v = D$) $m = 1 + \frac{D}{f}$

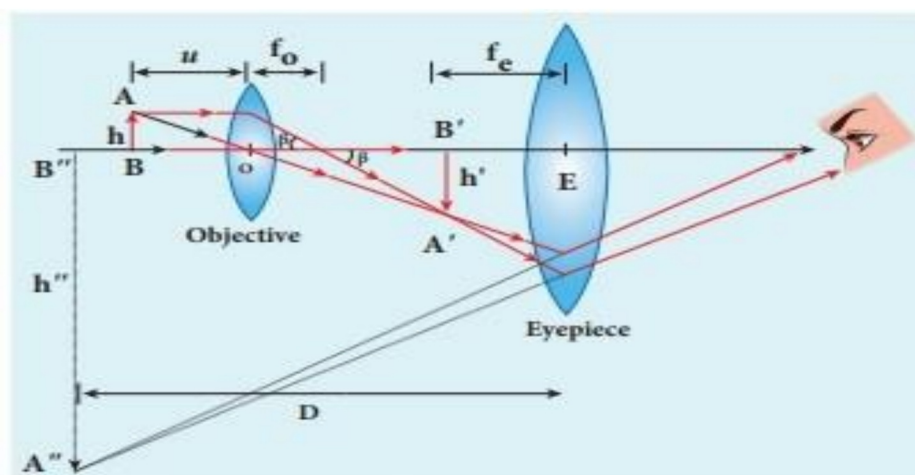
For image at far point ($v = \infty$) $m = \frac{D}{f}$



COMPOUND MICROSCOPE

Objective is a convex lens of short focal length and of small aperture.

Eye piece is a convex lens of slightly larger focal length of larger aperture.



Magnification of the compound microscope

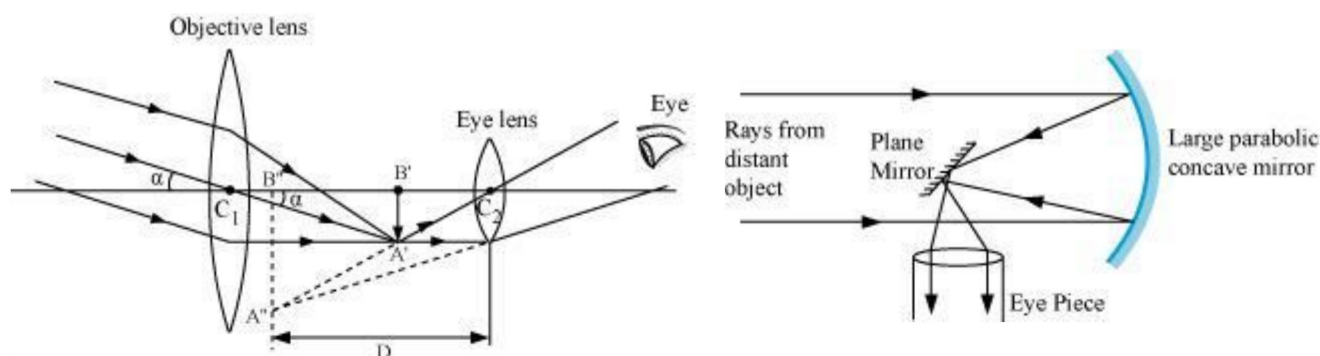
(a) when the final image is formed at the least distance of distinct vision is given by, $v_e = D$

$$m = -\frac{L}{f_o} \times \frac{D}{u_e} = m_o m_e = -\frac{v_o}{u_o} \left(1 + \frac{D}{f_e}\right)$$

(b) Magnification of the compound microscope when the final image is formed at the infinity is given by, $v_e = \infty$

$$m = m_o m_e = -\frac{v_o}{u_o} \left(\frac{D}{f_e}\right)$$

ASTRONOMICAL TELESCOPE



Magnifying power of telescope (m) The magnifying power of a telescope in normal adjustment is defined as the ratio of the angle subtended by the image at the eye / lens as seen through the telescope to the angle subtended by the object seen directly,

$$\text{Magnifying Power } m = \frac{\beta}{\alpha} = - \frac{AB/u_e}{AB/f_o} = - \frac{f_o}{u_e}$$

Magnification of the refracting telescope

Magnification of the reflecting telescope

- (i) when the final image is formed at the infinity is given by

$$m = - \frac{f_o}{f_e} \qquad m = \frac{f_o}{f_e} \left(1 + \frac{f_e}{D} \right)$$

- (ii) when the final image is formed at the infinity is given by

$$m = m_o m_e = - \frac{f_o}{f_e} \left(1 + \frac{f_e}{D} \right)$$

Advantages of reflecting type telescope over a refracting type telescope

1. No chromatic aberration because mirror is used
2. Spherical aberration gets removed by using a paraboloid mirror
3. The image is bright, because there is no loss of light due to reflection and absorption by objective
4. Higher resolution can be obtained by using a mirror of large aperture.
5. A mirror provides an easier mechanical support over its entire back surface.
6. It is difficult and expensive to make large sized lens free from chromatic aberrations and distortions